

Hybrid Model Predictive and Nonlinear Feedback Control for Chaos Suppression and MPPT Enhancement in DFIG-Based Wind Energy Systems

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Abstract—Doubly fed induction generator (DFIG)-based wind turbines are strongly nonlinear electromechanical systems that may exhibit irregular, and even chaotic, oscillations under abnormal operating conditions. This paper revisits the third-order rotor-current/speed model of the DFIG with a physically consistent parameter set, derives an equivalent dimensionless form, and shows by Lyapunov exponents and bifurcation analysis that chaotic motion appears only for strongly detuned mechanical damping and abnormal rotor excitation. A hybrid controller acting only on the rotor voltages is then proposed. An outer speed loop generates the torque-current reference, nonlinear feedback linearization cancels the coupled current dynamics, and a constrained, receding-horizon model predictive controller (MPC) with integral action handles the converter voltage limits; the MPC terminal weight and local gain are obtained from a linear matrix inequality (LMI) that guarantees nominal closed-loop stability. The controller is evaluated on chaos suppression and on maximum power point tracking (MPPT) of a 2-MW-class DFIG under turbulent wind and $\pm 20\%$ parameter mismatch, and compared with a feedback-linearization-only law and a PI law using the same gains. Using only the rotor voltages, with a cascaded speed loop, it suppresses chaos without steady-state error in all mismatch cases and settles 35–43 % faster than PI. It tracks the MPPT references with a speed error 5–140 times lower than PI and a power root-mean-square error of 0.6–1.7 % of rated power, three to eight times lower than with PI.

Index Terms—Doubly fed induction generator (DFIG), wind energy, chaotic dynamics, maximum power point tracking (MPPT), model predictive control (MPC), feedback linearization, linear matrix inequality (LMI).

I. INTRODUCTION

Rising electricity demand and the depletion of fossil resources call for sustainable energy sources such as wind [1]–[3]. Wind power has grown rapidly [4]: a record 117 GW was installed in 2024, bringing the cumulative capacity to about 1136 GW [5], driven by supportive policies and declining costs [6].

The doubly fed induction generator (DFIG) dominates large onshore wind farms because its partial-scale back-to-back converter, rated at about 25–30 % of the generator power, allows decoupled control of active and reactive power over a wide speed range [7]–[11]. Its dynamic modelling, vector control and fault ride-through have been studied extensively

[12]–[16], and grid codes impose strict requirements on its dynamic behaviour [17]–[19].

The electromechanical coupling between the rotor currents, the stator flux and the shaft speed makes the DFIG a strongly nonlinear system. Oscillatory transients [20] and, under particular parameter ranges or working conditions, chaotic behaviour have been reported [21], [22]; chaotic oscillations have also been analysed in other wind-turbine subsystems [23]. Chaos in electrical machines is usually analysed on dimensionless models [24], [25] and suppressed by nonlinear or adaptive feedback [24], [26], [27] or by predictive control [25], [28]–[31]; robust sliding-mode and fault-tolerant controllers have also been proposed for wind turbines [32], [33], and chaos synchronization has applications such as secure communications [34]. For MPPT, combining feedback linearization with MPC has proven effective [35].

A critical point, often overlooked, is whether the parameters used to exhibit chaos are physically admissible. In this paper: (i) the DFIG model of [36] is written with a physically consistent parameter set ($L_s, L_r > L_m$, $0 < \sigma < 1$) and all constants of the state equations are given explicitly; (ii) a dimensionless form is derived, which shows that chaos requires strongly abnormal damping and rotor excitation, and the chaotic regime is characterized by Lyapunov exponents and a bifurcation diagram; (iii) a hybrid controller combining feedback linearization with a constrained MPC is designed, with an LMI-based terminal cost that guarantees nominal stability; (iv) quantitative comparisons with two baselines under parameter mismatch are reported for chaos suppression and for MPPT tracking.

II. WIND TURBINE MODEL

The aerodynamic power captured by the rotor is [32], [33], [37]

$$P_a = \frac{1}{2} \rho \pi R^2 C_p(\lambda) v^3, \quad \lambda = \frac{R \Omega_t}{v}, \quad (1)$$

where ρ is the air density, R the blade radius, v the wind speed, Ω_t the turbine speed, and C_p the power coefficient, maximal ($C_{p,\max}$) at $\lambda = \lambda_{opt}$. We use $C_p(\lambda) = 0.5176(116/\lambda_i - 5)e^{-21/\lambda_i} + 0.0068\lambda$ with $1/\lambda_i = 1/\lambda - 0.035$ (zero pitch).

Through a gearbox of ratio G , the generator mechanical speed is $\Omega_m = G\Omega_t$ and the electrical rotor speed is $\omega_r = n_p\Omega_m$, n_p being the number of pole pairs. With the motor sign convention, the shaft dynamics referred to the generator side are

$$J \frac{d\Omega_m}{dt} = T_e - T_L - D\Omega_m, \quad T_L = -\frac{P_a}{\Omega_m}, \quad (2)$$

where J is the total equivalent inertia, D the viscous damping, and T_e the electromagnetic torque ($T_e < 0$ in generating mode). In the synchronous dq frame, $T_e = \frac{3}{2}n_p(\psi_{ds}i_{qs} - \psi_{qs}i_{ds})$.

III. DFIG MODEL AND CHAOTIC DYNAMICS

A. Physically Consistent State Model

The stator flux is aligned with the q axis ($\psi_{qs} = \psi_s$, $\psi_{ds} = 0$) and, neglecting the stator resistance in the flux relation, $u_{ds} = -V_s$, $u_{qs} = 0$, $\psi_s = V_s/\omega_s$, where V_s is the stator-voltage amplitude and ω_s the grid angular frequency. Then $i_{ds} = -(L_m/L_s)i_{dr}$ and $T_e = \frac{3}{2}n_p(L_m/L_s)\psi_s i_{dr}$. With $x = [x_1, x_2, x_3]^T = [i_{dr}, i_{qr}, \omega_r]^T$, the DFIG dynamics [36] read

$$\begin{cases} \dot{x}_1 = a_1x_1 + (\omega_s - x_3)x_2 + a_2x_3 + c_1 + u_1, \\ \dot{x}_2 = -(\omega_s - x_3)x_1 + a_1x_2 + c_2 + u_2, \\ \dot{x}_3 = a_6x_1 - a_7x_3 - a_8, \end{cases} \quad (3)$$

with $\sigma = 1 - L_m^2/(L_sL_r)$ and

$$\begin{aligned} a_1 &= -\frac{R_sL_m^2}{\sigma L_s^2L_r} - \frac{R_r}{\sigma L_r}, & a_2 &= -\frac{L_m\psi_s}{\sigma L_rL_s}, \\ a_3 &= -\frac{L_m}{\sigma L_sL_r}, & a_4 &= \frac{1}{\sigma L_r}, & a_5 &= \frac{R_sL_m\psi_s}{\sigma L_s^2L_r}, \\ a_6 &= \frac{3n_p^2L_m\psi_s}{2JL_s}, & a_7 &= \frac{D}{J}, & a_8 &= \frac{n_p}{J}T_L, \\ c_1 &= a_3u_{ds} + a_4u_{dr}^0, & c_2 &= a_5 + a_3u_{qs} + a_4u_{qr}^0. \end{aligned} \quad (4)$$

Here u_{dr}^0, u_{qr}^0 are constant (open-loop) rotor voltages and $u_{1,2} = a_4\Delta u_{dr,qr}$ are the control actions of the rotor-side converter, which are the only available inputs: the speed equation has no actuator and is governed through the electromagnetic torque, i.e., through i_{dr} . Without control, $u_1 = u_2 = 0$.

Table I lists the nominal parameters of a representative 2-MW-class DFIG (stator-referred). They satisfy $L_s, L_r > L_m$ and $\sigma = 0.0661 \in (0, 1)$. The resulting coefficients are $a_1 = -31.14 \text{ s}^{-1}$, $a_2 = -1.013 \times 10^4 \text{ A/rad}$, $a_3 = -5.649 \times 10^3$ and $a_4 = 5.845 \times 10^3 \text{ A/(V}\cdot\text{s)}$, $a_5 = 1.018 \times 10^4 \text{ A/s}$, $a_6 = 8.19 \times 10^{-2} \text{ rad/(A}\cdot\text{s}^2)$ and $a_7 = 7.87 \times 10^{-6} \text{ s}^{-1}$. Because $u_{ds} = -\omega_s\psi_s$, $c_1 + a_2\omega_s = a_4u_{dr}^0$: the constant $a_3u_{ds} = 3.18 \times 10^6 \text{ A/s}$ exactly compensates $a_2\omega_s$, and $c_2 = 1.018 \times 10^4 + a_4u_{qr}^0 \text{ A/s}$. Every constant of (3) is therefore obtained from Table I.

TABLE I
NOMINAL PARAMETERS (2-MW-CLASS DFIG AND TURBINE)

Param.	Value	Param.	Value
R_s	2.6 m Ω	n_p	2
R_r	2.9 m Ω	V_s (line, rms)	690 V, 50 Hz
L_m	2.5 mH	ψ_s	1.793 Wb
$L_s = L_r$	2.587 mH	J	127 kg·m ²
σ	0.0661	D	10 ⁻³ N·m·s/rad
$\tau = -1/a_1$	32.1 ms	R, G	40 m, 80
ρ	1.225 kg/m ³	$\lambda_{opt}, C_{p,max}$	8.1, 0.48

B. Dimensionless Model and Conditions for Chaos

Let $\tau = -1/a_1$, $\tilde{t} = t/\tau$, $k = a_7/(\tau a_6)$, and $z_1 = x_1/k$, $z_2 = x_2/k$, $z_3 = \tau(x_3 - \omega_s)$ (normalized slip speed). Then (3) with $u = 0$ becomes

$$\begin{cases} \dot{z}_1' = -z_1 - z_3z_2 + \gamma z_3 + \varepsilon_1, \\ \dot{z}_2' = -z_2 + z_3z_1 + \varepsilon_2, \\ \dot{z}_3' = s(z_1 - z_3) + \varepsilon_3, \end{cases} \quad (5)$$

where $(\cdot)' = d/d\tilde{t}$, $s = \tau D/J$, $\varepsilon_1 = \tau(c_1 + a_2\omega_s)/k$, $\varepsilon_2 = \tau c_2/k$, $\varepsilon_3 = -\tau^2(a_7\omega_s + a_8)$, and

$$\gamma = \frac{a_2a_6\tau}{a_7} = -\frac{3n_p^2L_m^2\psi_s^2\tau}{2\sigma L_rL_s^2D} < 0. \quad (6)$$

With physical parameters ($\sigma > 0$), γ is always negative. The divergence of (5) is $-(2+s) < 0$, so the system is dissipative.

For the nominal machine, $s = 2.5 \times 10^{-7}$ and the equilibrium is stable. Chaos appears only far from nominal. For $\gamma = -1$, $s = 8$, $\varepsilon_1 = \varepsilon_3 = 0$ and $\varepsilon_2 = -85$, the Lyapunov spectrum, computed with the QR-based variational method [38] over 500 time units, is (1.416, 0.003, -11.419). Its sum equals the divergence -10, and the Kaplan-Yorke dimension is 2.12. Fig. 1(a)–(b) shows the resulting double-scroll attractor. The bifurcation diagram and the largest Lyapunov exponent versus ε_2 (Fig. 1(c)–(d)) show a stable equilibrium for $\varepsilon_2 > -15$ and chaos for $\varepsilon_2 < -15$, interrupted by periodic windows near $\varepsilon_2 \approx -50.5$, -41.8 and -28.5.

Mapped back to Table I, this chaotic set requires $D/J \approx 250 \text{ s}^{-1}$ (e.g., $J \approx 13.6 \text{ kg}\cdot\text{m}^2$ and $D \approx 3.4 \text{ kN}\cdot\text{m}\cdot\text{s/rad}$) and a q -axis rotor voltage of several kilovolts. Chaos is therefore not a feature of normal operation of this machine but a severe fault/detuning scenario, for which the controller must still guarantee recovery. Following common practice [24], [25], chaos suppression is studied on (5), whereas MPPT tracking is studied in physical units.

IV. MPPT REFERENCE GENERATION

The wind speed is modelled as a mean value, three low-frequency harmonics and a smoothed Gaussian turbulence v_{turb} (first-order filter, time constant 0.5 s):

$$v(t) = V_{mean} + \sum_{k=1}^3 A_k \sin(2\pi f_k t + \phi_k) + v_{turb}(t), \quad (7)$$

with $V_{mean} = 9 \text{ m/s}$, $(A_k) = (1.3, 0.7, 0.4) \text{ m/s}$ and $(f_k) = (0.05, 0.13, 0.31) \text{ Hz}$ (Fig. 2).

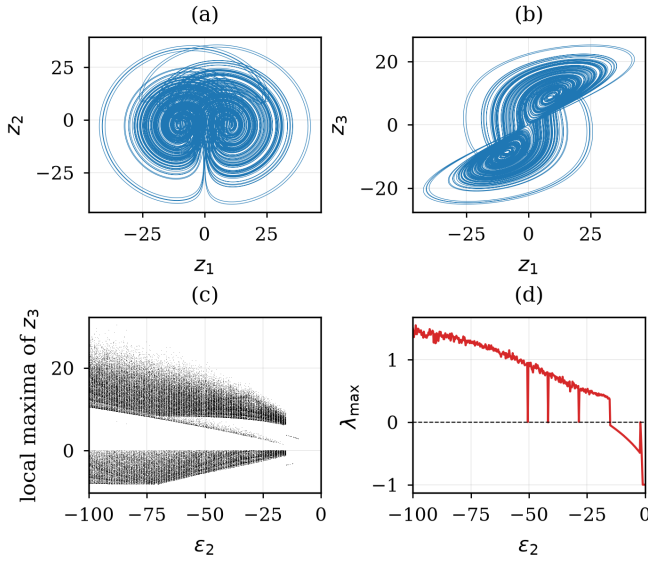


Fig. 1. Chaotic dynamics of (5) ($\gamma = -1$, $s = 8$, $\varepsilon_1 = \varepsilon_3 = 0$): (a) z_1 - z_2 and (b) z_1 - z_3 projections for $\varepsilon_2 = -85$; (c) bifurcation diagram (local maxima of z_3) and (d) largest Lyapunov exponent versus ε_2 .

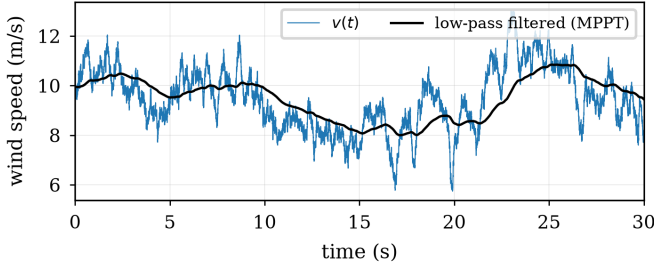


Fig. 2. Turbulent wind-speed profile and its low-pass filtered version used by the MPPT.

The optimal speed follows the low-pass filtered wind $\bar{v}$ (time constant 2 s):

$$\omega_{r,ref}(t) = n_p G \lambda_{opt} \bar{v}(t) / R. \quad (8)$$

Since $T_e \propto i_{dr}$, the d -axis current reference is obtained from the torque balance of (3), so that it is consistent with $\omega_{r,ref}$:

$$i_{dr,ref} = \frac{\dot{\omega}_{r,ref} + a_7 \omega_{r,ref} + (n_p / J) T_L(v, \omega_{r,ref})}{a_6}. \quad (9)$$

The stator reactive power is $Q_s = \frac{3}{2} V_s i_{qs} = \frac{3}{2} V_s (\psi_s - L_m i_{qr}) / L_s$, so unity power factor ($Q_s = 0$) gives

$$i_{qr,ref} = \psi_s / L_m. \quad (10)$$

The generated stator active power and its reference are

$$P_s = -\frac{3}{2} V_s \frac{L_m}{L_s} i_{dr}, \quad P_{s,ref} = -\frac{3}{2} V_s \frac{L_m}{L_s} i_{dr,ref}. \quad (11)$$

The reference trajectory is $x_d = [i_{dr,ref}, i_{qr,ref}, \omega_{r,ref}]^T$ and the tracking error is $e = x - x_d$.

V. HYBRID FEEDBACK-LINEARIZATION/MPC DESIGN

A. Outer Speed Loop

Since $T_e \propto i_{dr}$, the speed is controlled through the reference of i_{dr} (cascade structure). With $e_\omega = \omega_r - \omega_{r,ref}$ and the nominal model,

$$\dot{i}_{dr}^* = \frac{\dot{\omega}_{r,ref} + a_7 \omega_r + \hat{a}_8(v, \omega_r) - \lambda e_\omega - \lambda_I \int e_\omega dt}{a_6}, \quad (12)$$

which reduces to (9) when $e_\omega = 0$. Substituting into the third line of (3) gives

$$\dot{e}_\omega = -\lambda e_\omega - \lambda_I \int e_\omega dt + a_6 (i_{dr} - i_{dr}^*).$$

This is an exponentially stable linear system (poles: roots of $p^2 + \lambda p + \lambda_I$) driven by the inner tracking error. It is therefore input-to-state stable, and $e_\omega \rightarrow 0$ once $i_{dr} \rightarrow i_{dr}^*$ (cascade argument). The inner reference is $x_d = [i_{dr}^*, i_{qr,ref}]^T$, and $\dot{x}_d$ is computed analytically from (3) and (12).

B. Inner Loop: Feedback Linearization

Write the first two lines of (3) as $\dot{x}_{1,2} = F_{1,2}(x, t) + u_{1,2}$ and let $\hat{F}_{1,2}$ be their nominal model. The control law

$$u_{1,2} = -\hat{F}_{1,2}(x, t) + \dot{x}_d + v \quad (13)$$

gives the error dynamics $\dot{e} = v + \Delta(x, t)$, $e = [i_{dr} - i_{dr}^*, i_{qr} - i_{qr,ref}]^T$, with $\Delta = F_{1,2} - \hat{F}_{1,2}$ ($\Delta = 0$ nominally). This cancels the bilinear couplings $(\omega_s - x_3)x_{1,2}$ responsible for chaos, and leaves two decoupled integrators driven by the new input v .

C. Constrained MPC with Integral Action

For each channel i , the error is augmented with its integral, $\xi_i = [e_i, \int e_i dt]^T$, to remove steady-state errors due to Δ . Sampling with period T_s gives

$$\xi_i(k+1) = A \xi_i(k) + B v_i(k), \quad A = \begin{bmatrix} 1 & 0 \\ T_s & 1 \end{bmatrix}, \quad B = \begin{bmatrix} T_s \\ T_s^2/2 \end{bmatrix}. \quad (14)$$

At each instant k , the MPC solves the quadratic program

$$\begin{aligned} \min_{v_i(\cdot|k)} \quad & \sum_{j=1}^N \|\xi_i(k+j|k)\|_Q^2 + \|\xi_i(k+N|k)\|_P^2 \\ & + r \sum_{j=0}^{N-1} v_i(k+j|k)^2 \\ \text{s.t.} \quad & (14), \quad \underline{v}_i(k) \leq v_i(k+j|k) \leq \bar{v}_i(k), \end{aligned} \quad (15)$$

where $Q = Q^T > 0$, $r > 0$, and the input bounds $\underline{v}_i = -\bar{u}_i - h_i$, $\bar{v}_i = \bar{u}_i - h_i$, with $h = -\hat{F} + \dot{x}_d$, enforce the actuator limit $|u_i| \leq \bar{u}_i$ on (13). Only the first optimal move is applied (receding horizon). The integral state is frozen when the limit is reached (conditional integration).

Theorem 1: Let $X = X^T > 0$ and Y satisfy the LMI

$$\begin{bmatrix} X & (AX + BY)^T & XQ^{1/2} & \sqrt{r}Y^T \\ AX + BY & X & 0 & 0 \\ Q^{1/2}X & 0 & I & 0 \\ \sqrt{r}Y & 0 & 0 & 1 \end{bmatrix} \succeq 0, \quad (16)$$

and set $P = X^{-1}$, $K = YX^{-1}$. Then

$$(A + BK)^T P (A + BK) - P + Q + rK^T K \preceq 0. \quad (17)$$

Let $\Omega = \{\xi : \xi^T P \xi \leq \alpha\}$, with $\alpha > 0$ small enough that $K\xi$ satisfies the input bounds in Ω . If $\Delta = 0$ and (15), augmented with the terminal constraint $\xi_i(k+N|k) \in \Omega$, is feasible at $k = 0$, then it remains feasible for all k and $\xi_i = 0$ is asymptotically stable, i.e., $x_{1,2} \rightarrow x_d$ and, with the outer loop, $\omega_r \rightarrow \omega_{r,ref}$.

Proof: Applying the Schur complement to (16) with respect to its last three block rows gives $X - (AX + BY)^T X^{-1} (AX + BY) - XQX - rY^T Y \succeq 0$. Pre- and post-multiplying by $P = X^{-1}$ and substituting $Y = KX$ yields (17). Hence Ω is positively invariant under $v = K\xi$, and $\xi^T P \xi$ is a control Lyapunov function in Ω . By the standard terminal-cost/terminal-set argument [39], the optimal cost V_N^* of (15) then satisfies $V_N^*(k+1) - V_N^*(k) \leq -\|\xi_i(k)\|_Q^2 - r v_i(k)^2$ and the shifted solution completed by $K\xi$ is feasible, which proves recursive feasibility and asymptotic stability. ■

Remarks: (i) The maximal-volume solution of (16) satisfies (17) with equality, i.e., P is the solution of the discrete algebraic Riccati equation [40]. Consequently, when no constraint is active, (15) returns exactly $v_i = K\xi_i$. A single gain $K = [k_e, k_I]$, common to the two channels, is therefore obtained from the LMI and used in the implementation. (ii) Setting $k_I = 0$ and ignoring constraints recovers the pure feedback-linearization law $v = -k_e e$ (“FL only”). Removing all model terms gives the classical cascaded PI vector control: a speed PI $i_{dr}^* = -(\lambda e_\omega + \lambda_I \int e_\omega)/a_6$ and current PIs $v = K\xi$. These two baselines use the same gains as the proposed controller, so the comparison isolates the effect of feedback linearization and of constrained MPC. (iii) The terminal constraint is not imposed explicitly in the simulations; convergence was verified a posteriori.

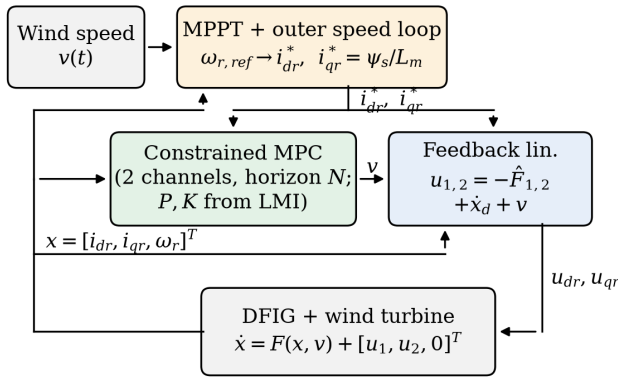


Fig. 3. Proposed cascaded architecture: outer speed loop, inner feedback-linearization/MPC current loops acting on the rotor voltages.

The overall structure is shown in Fig. 3. The QP (15) has only N variables per channel and was solved by a bounded-variable least-squares method.

VI. SIMULATION RESULTS

All simulations integrate the plant with a fourth-order Runge–Kutta scheme (4–5 sub-steps per sampling period). The

controller uses the nominal model; the plant uses the nominal or perturbed parameters.

A. Chaos Suppression

System (5) is started in the chaotic regime and the controller is switched on at $\tilde{t}_{on} = 6$ with $T_s = 0.01$, $N = 15$, $Q = \text{diag}(1, 10)$, $r = 10^{-4}$ and $|u_{1,2}| \leq 120$ (no input on the speed equation). The LMI gives $K = [-63.9, -192.4]$, and the outer loop on z_3 uses $\lambda = 10$, $\lambda_I = 25$ (double pole at -5). The target $z^* = (2.02, -2.00, 2.02)$ corresponds to $\omega_r = 1.2\omega_s$ (super-synchronous operation with positive speed). It is an equilibrium with $u_1^* = 0$, sustained by $u_2^* = 78.9$ alone, which compensates the abnormal excitation ε_2 . Robustness is tested by scaling all coefficients of the plant by 0.8 and 1.2.

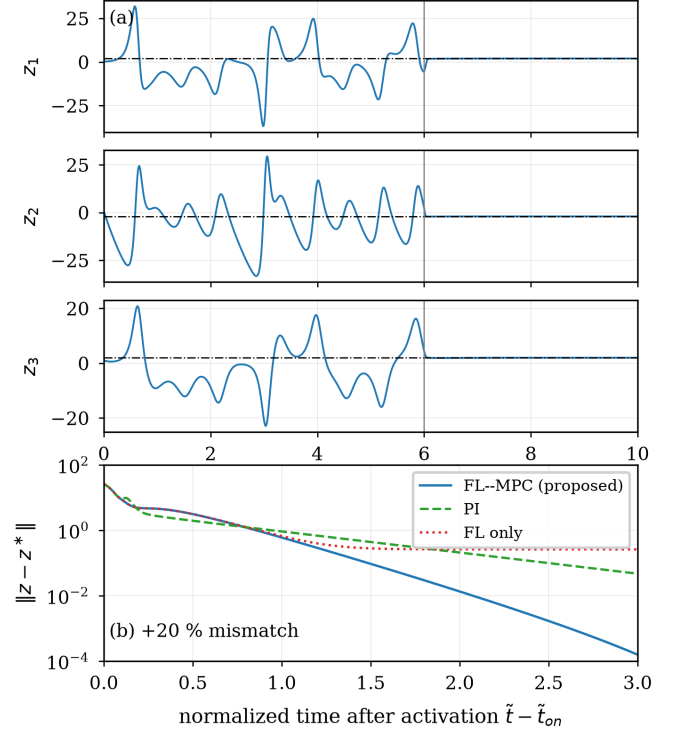


Fig. 4. Chaos suppression: (a) states of (5) with FL-MPC switched on at $\tilde{t} = 6$ (dash-dot: target z^*); (b) error norm after activation with +20 % plant mismatch.

TABLE II
CHAOS SUPPRESSION PERFORMANCE WITH u_1, u_2 ONLY (IAE OF $\|z - z^*\|$, SETTLING TIME t_s TO A 2 % BAND, FINAL ERROR e_f ; NORMALIZED TIME)

Plant	Metric	FL-MPC	FL only	PI
nominal	IAE	0.42	0.41	0.72
	t_s	0.66	0.60	1.01
	e_f	0	0	0
−20 %	IAE	2.18	4.51	1.98
	t_s	1.30	–	2.13
	e_f	0	0.26	0
+20 %	IAE	4.55	6.81	4.56
	t_s	1.57	–	2.73
	e_f	0	0.26	0

Fig. 4(a) shows that the chaotic oscillations are suppressed as soon as the controller is activated, using only the rotor voltages, and the states settle at z^* . Table II and Fig. 4(b) compare the three controllers. With an exact model, FL-MPC and FL only behave almost identically. Under mismatch, FL only keeps a steady-state error (0.26). PI and FL-MPC both converge without offset and have similar IAE, but FL-MPC settles 35–43 % faster (0.66–1.57 versus 1.01–2.73, i.e., 21–50 ms versus 32–88 ms with $\tau = 32.1$ ms). All controllers reach the input limit during the first instants; the MPC accounts for it explicitly.

B. MPPT Tracking of the 2-MW-Class DFIG

The model (3) with Table I is driven by the wind of Fig. 2 for 30 s, with $T_s = 1$ ms, $N = 10$, $Q = \text{diag}(1, 3)$ and $r = 10^{-6}$ (errors scaled by 1 kA), and the outer speed loop uses $\lambda = 10 \text{ s}^{-1}$, $\lambda_I = 25 \text{ s}^{-2}$. The only inputs are the rotor voltages, limited to $|\Delta u_{dr}|, |\Delta u_{qr}| \leq 200$ V. Two mismatched plants are considered: M1 ($R_s, R_r + 20\%$, $L_m - 10\%$, $J + 20\%$) and M2 ($R_s, R_r - 20\%$, $L_m + 10\%$, $J - 20\%$), with the leakage inductances kept constant so that $L_s, L_r > L_m$ still holds.

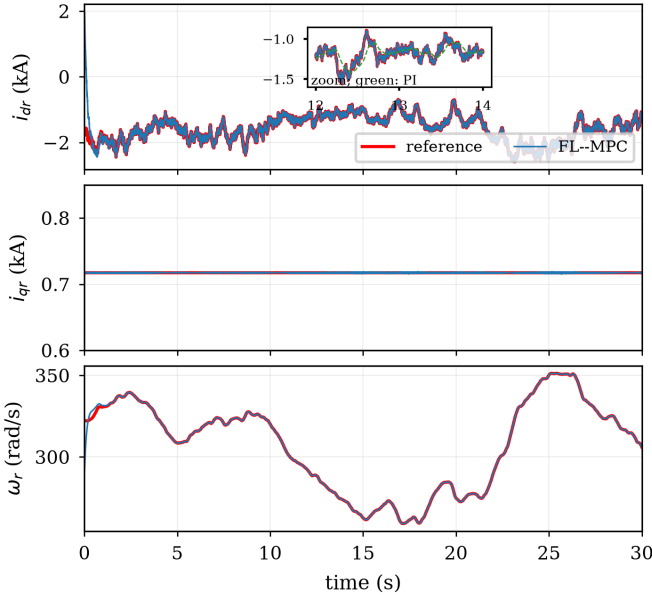


Fig. 5. Tracking of i_{dr} , i_{qr} and ω_r with FL-MPC (nominal plant). Inset: zoom including PI.

Figs. 5 and 6 show that the rotor currents, the speed and the stator power follow the MPPT references despite turbulence. The speed stays in the admissible slip range ($0.82\text{--}1.12 \omega_s$), and P_s varies between 0.57 and 2.1 MW. Starting from $0.9 \omega_{r,ref}$, the speed enters the 2 % band in 0.13 s with FL-MPC, versus 0.77 s with PI. Table III shows that the speed RMS error of FL-MPC is 5–140 times lower than with PI, and its power RMS error is 0.59 % of rated power (nominal) and 1.65 % (mismatch), about three to eight times lower than with PI. Under mismatch, the larger i_{dr} and P_s errors with respect to the nominal MPPT reference mainly reflect that

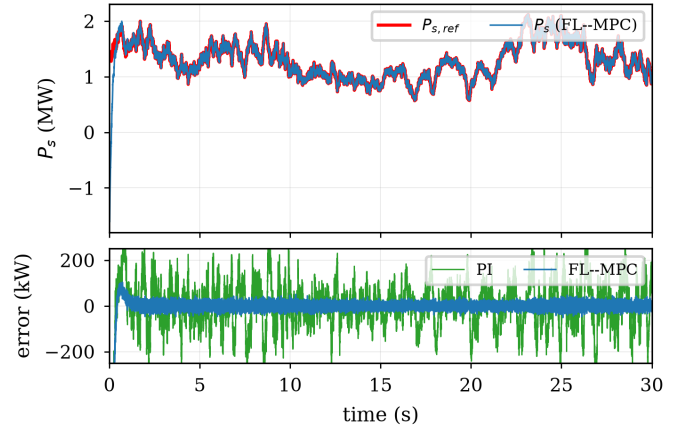


Fig. 6. Stator active power P_s and MPPT reference $P_{s,ref}$ (FL-MPC), and power tracking errors of FL-MPC and PI.

TABLE III
MPPT TRACKING PERFORMANCE ($t \geq 5$ s): RMS ERRORS OF P_s (% OF 2 MW), i_{dr} (A) AND ω_r (RAD/S)

Plant	RMSE	FL-MPC	FL only	PI
nominal	P_s	0.59	0.59	4.61
	i_{dr}	14.3	14.3	112.9
	ω_r	0.008	0.008	1.157
M1	P_s	1.65	1.65	5.02
	i_{dr}	40.5	40.5	122.8
	ω_r	0.184	0.184	0.973
M2	P_s	1.65	1.65	4.59
	i_{dr}	40.3	40.3	112.4
	ω_r	0.183	0.183	1.345

the torque current actually required changes with J and L_m , while the speed is still tracked within 0.18 rad/s. For tracking, FL-MPC and FL only perform identically: the outer integral action compensates the mismatch, and the voltage limits are never reached by FL-MPC in this scenario (maximum 185 V). The main benefits of the MPC are therefore its explicit, stability-preserving treatment of constraints and its offset-free behaviour, which matter in large transients such as the recovery from chaos.

VII. CONCLUSION

This paper revisited the nonlinear dynamics of a DFIG-based wind turbine with a physically consistent parameter set and explicit model constants. A dimensionless model shows that chaos, confirmed by a positive Lyapunov exponent and a bifurcation diagram, arises only under severely detuned damping and abnormal rotor excitation, i.e., as a fault scenario. A hybrid controller combining feedback linearization with a constrained MPC whose terminal ingredients are obtained from an LMI was proposed, with guaranteed nominal stability. The control acts only on the rotor voltages, the speed being governed through a cascaded torque-current loop. Simulations show chaos suppression without steady-state error under $\pm 20\%$ mismatch with faster settling than PI, and MPPT tracking with speed and power errors several times lower than

PI. Future work will address converter and grid dynamics, low-voltage ride-through, and experimental validation.

CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest.

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